



Zain's Serenity – Luxury Coastal Retreat

A full-stack luxury resort website built with Next.js 16, MongoDB, and modern web technologies



Live Website: zainserenity.vercel.app



Screenshots

Hero Section	Featured Rooms
Hero	Featured Rooms
Rooms Page	Room Detail
Rooms	Room Management
Admin Dashboard	Admin Bookings
Dashboard	Bookings
Gallery Management	User Management
Gallery	Users
Booking Email	Guest Review
Email	Review
My Bookings	Why Choose Us
My Bookings	Why Us

GA4 Tracking	Contact Form
 GA4	 Contact
Gallery Page	Add Room
 Gallery	 Add Room

Overview

Zain's Serenity is a complete luxury coastal retreat website that offers a seamless booking experience for guests while providing a powerful admin dashboard for managing rooms, bookings, users, gallery images, and analytics. Built with cutting-edge technologies, the website delivers a premium user experience with smooth animations, 3D room tours, and real-time availability management.

The project represents the vision of **Zain Shah** — a sanctuary where nature, culture, and luxury coexist. From the initial discovery of Crystal Cove in 2009 to a world-class resort recognized among the **Top 10 Luxury Resorts Worldwide** in 2025, this digital platform brings that dream to life.

Key Features

Guest Features

Feature	Description
Dynamic Room Catalog	Browse rooms with pagination, featured badges, ratings, and real-time availability filters
Advanced Search	Search by check-in/check-out dates with automatic availability filtering
Room Detail View	Full gallery, amenities, reviews, availability calendar, and 3D room viewer

Feature	Description
Booking System	Complete reservation flow with guest/children count, special requests, and instant price calculation
User Authentication	Email/password registration with verification, Google OAuth, password reset, and session management
My Bookings	View, filter, and track all personal bookings with status indicators
Review System	Submit ratings and reviews for rooms with duplicate prevention
Gallery	Category-filtered image gallery with lightbox and dynamic loading
Contact Form	Professional inquiry form with email notifications
Interactive Maps	Embedded location with coordinates and directions link

Admin Features

Feature	Description
Analytics Dashboard	Real-time revenue charts, booking status, occupancy rates, and room performance
Revenue Tracking	Custom date ranges, room-wise breakdown, monthly comparisons
Booking Management	Confirm/cancel bookings with automated email notifications
Room Management	CRUD operations, image uploads, featured/new flags, status toggles
Gallery Management	Upload, categorize, caption, and delete images with Cloudinary integration
User Management	Role assignment, verification toggle, user deletion with confirmation
Availability Grid	14-day visual room availability matrix
Notification System	Real-time alerts for pending bookings and today's check-ins



Advanced Integrations

Integration	Purpose
Google Analytics 4	Complete event tracking for user behavior, bookings, and engagement
Cloudinary	Cloud-based image storage and optimization
Nodemailer	Professional HTML email templates for booking confirmations, verification, and password resets
Three.js / React Three Fiber	Interactive 3D bedroom model with auto-rotation
Recharts	Beautiful data visualizations for admin analytics
Framer Motion	Smooth animations and transitions throughout the site



Tech Stack

Frontend

- **Next.js 16.3.1** — React framework with App Router
- **React 19.2.8** — UI library
- **TypeScript 5** — Type safety and better developer experience
- **Tailwind CSS 4** — Utility-first CSS framework
- **Framer Motion 13** — Animation library
- **Lucide React** — Icon library
- **React DatePicker** — Date selection components

Backend

- **Next.js API Routes** — Serverless API endpoints
- **MongoDB 9.9.2** — NoSQL database
- **Mongoose** — ODM for MongoDB
- **NextAuth.js 4.24** — Authentication
- **Bcrypt.js** — Password hashing
- **Nodemailer 7** — Email sending

3D & Visualization

- **Three.js 0.185** — 3D graphics
- **React Three Fiber 9.7** — React renderer for Three.js
- **React Three Drei 10.7** — Useful helpers for R3F

DevOps & Deployment

- **Vercel** — Hosting and deployment
- **Cloudinary** — Image storage
- **Google Analytics 4** — Analytics
- **GitHub** — Version control



Installation

Prerequisites

- **Node.js 18+** (recommended: 20+)
- **npm** or **yarn** or **pnpm**
- **MongoDB** — local or [MongoDB Atlas](#)
- **Cloudinary Account** — for image uploads
- **Google OAuth Credentials** — for social login
- **Gmail Account** — for sending emails (with app password)

Step 1: Clone the Repository

```
git clone https://github.com/zainshah3464/Zain-Serenity.git
cd Zain-Serenity
```

Step 2: Install Dependencies

```
npm install
# or
yarn install
# or
pnpm install
```

Step 3: Configure Environment Variables

Create a `.env.local` file in the root directory:

```
# — Database —————
MONGODB_URI=mongodb+srv://your-username:your-password@cluster.mongodb.net/guesthouse

# — Authentication —————
NEXTAUTH_SECRET=your-nextauth-secret-key
NEXTAUTH_URL=http://localhost:3000

# — Google OAuth —————
GOOGLE_CLIENT_ID=your-google-client-id
GOOGLE_CLIENT_SECRET=your-google-client-secret

# — Cloudinary —————
CLOUDINARY_CLOUD_NAME=your-cloud-name
CLOUDINARY_API_KEY=your-api-key
CLOUDINARY_API_SECRET=your-api-secret

# — Email (Gmail) —————
EMAIL_USER=your-email@gmail.com
EMAIL_PASS=your-gmail-app-password

# — Google Analytics 4 —————
NEXT_PUBLIC_GA_ID=G-XXXXXXXXXX

# — Base URL —————
NEXT_PUBLIC_BASE_URL=http://localhost:3000
```

Step 4: Run Development Server

```
npm run dev
```

Open <http://localhost:3000> in your browser.

Step 5: Build for Production

```
npm run build
npm start
```

Project Structure

```
Zain-Serenity/
├─ public/
│   └─ images/
│       └─ about/          # About page images
```

```

| | | └─ favicon/           # Favicon files & manifest
| | | └─ gallery/           # Gallery images
| | | └─ location/          # Location/map images
| | | └─ screenshots/       # Project screenshots
| | | └─ why-us/            # Why Choose Us images
| | | └─ email-header.png   # Email template header
| | | └─ hero-fallback.jpg   # Hero fallback image
| | |   └─ og-image.png      # Open Graph image
| | └─ models/
| |   └─ bedroom.glb         # 3D bedroom model
| └─ videos/
|   └─ hero-video.mp4        # Hero background video
|   └─ ...
|
└─ src/
  └─ app/                    # Next.js App Router pages
    └─ about/
    └─ admin/
      └─ (private)/
        └─ bookings/
        └─ gallery/
        └─ rooms/
        └─ [id]/
          └─ new/
          └─ users/
        └─ (public)/
          └─ login/
      └─ api/
        └─ admin/
          └─ bookings/[id]/
          └─ dashboard/
          └─ gallery/
          └─ rooms/
          └─ stats/
          └─ upload/
          └─ users/
          └─ verify-user/
        └─ auth/
          └─ [...nextauth]/
          └─ forgot-password/
          └─ register/
          └─ reset-password/
          └─ verify/

```

```
| | | └─ bookings/
| | | └─ contact/
| | | └─ gallery/
| | | └─ reviews/
| | | └─ rooms/
| | └─ booking/
| | └─ contact/
| | └─ forgot-password/
| | └─ gallery/
| | └─ login/
| | └─ my-bookings/
| | └─ register/
| | └─ reset-password/
| | └─ rooms/
| | └─ favicon.ico
| | └─ globals.css
| | └─ layout.tsx
| | └─ not-found.tsx
| | └─ page.tsx
|
| └─ components/
|   └─ admin/
|     └─ AdminDashboard.tsx
|     └─ AdminSidebar.tsx
|     └─ AdminTopbar.tsx
|     └─ DashboardComponents.tsx
|     └─ DashboardEntryAnimation.tsx
|     └─ DashboardLoader.tsx
|     └─ DetailModal.tsx
|   └─ AboutContent.tsx
|   └─ FeaturedRoomsCarousel.tsx
|   └─ Footer.tsx
|   └─ Gallery.tsx
|   └─ HeroLoadingContext.tsx
|   └─ LocationMap.tsx
|   └─ Navbar.tsx
|   └─ Providers.tsx
|   └─ PublicLayoutWrapper.tsx
|   └─ ReviewCarousel.tsx
|   └─ ReviewForm.tsx
|   └─ Room3DViewer.tsx
|   └─ RoomCard.tsx
|   └─ RoomDetailContent.tsx
```



```

|   |   |─ ScrollReveal.tsx
|   |   |─ searchBar.tsx
|   |   |─ SkeletonCard.tsx
|   |   |─ StoryHero.tsx
|   |   └─ WhyChooseUs.tsx
|   |
|   |─ hooks/
|   |   └─ useMediaQuery.ts
|   |
|   |─ lib/
|   |   └─ bookingValidation.ts
|   |   └─ checkAvailability.ts
|   |   └─ dbConnect.ts
|   |   └─ emailTemplates.ts
|   |   └─ ga4.ts
|   |   └─ rateLimiter.ts
|   |
|   |─ models/
|   |   └─ Booking.ts
|   |   └─ GalleryImage.ts
|   |   └─ Review.ts
|   |   └─ Room.ts
|   |   └─ User.ts
|   |
|   └─ types/
|       └─ next-auth.d.ts
|
└─ .env.local                # Environment variables (not in git)
└─ .gitignore
└─ AGENTS.md
└─ CLAUDE.md
└─ README.md
└─ eslint.config.mjs
└─ next-env.d.ts
└─ next.config.ts
└─ package.json
└─ postcss.config.mjs
└─ tailwind.config.js
└─ tsconfig.json

```



Database Models

User Model (src/models/User.ts)

Field	Type	Description
name	String	Full name of user
email	String	Unique email address
passwordHash	String?	Hashed password (optional for Google users)
role	"admin" "customer"	User role (default: "customer")
isVerified	Boolean	Email verification status
verificationToken	String?	Token for email verification
verificationTokenExpiry	Number?	Token expiry timestamp
resetToken	String?	Password reset token
resetTokenExpiry	Number?	Reset token expiry
lastLogin	Date?	Last login timestamp

Room Model (src/models/Room.ts)

Field	Type	Description
name	String	Room name
description	String	Detailed description
price	Number	Price per night
image	String	Main image URL
images	String[]	Multiple images
capacity	Number	Maximum guests

Field	Type	Description
amenities	String[]	List of amenities
status	"active" "inactive" "maintenance"	Room availability status
isFeatured	Boolean	Featured room flag
isNewRoom	Boolean	New room flag
view	String?	View type (e.g., "Mountain view")
rating	Number	Average rating (0-5)
roomType	String	Type (standard/deluxe/suite)
bedType	String	Bed configuration
size	String	Room size in sq ft
discount	Number?	Discount percentage

Booking Model (src/models/Booking.ts)

Field	Type	Description
userId	String	Reference to user
roomId	String	Reference to room
checkIn	Date	Check-in date
checkOut	Date	Check-out date
guests	Number	Number of guests
children	Number	Number of children
totalPrice	Number	Total booking price
specialRequests	String?	Special requests

Field	Type	Description
status	"pending" "confirmed" "cancelled"	Booking status
createdAt	Date	Creation timestamp

Review Model (src/models/Review.ts)

Field	Type	Description
userId	ObjectId	Reference to user
roomId	String	Reference to room
rating	Number	Rating (1-5)
comment	String	Review comment
createdAt	Date	Creation timestamp

GalleryImage Model (src/models/GalleryImage.ts)

Field	Type	Description
url	String	Image URL (Cloudinary)
caption	String?	Image caption
category	"featured" "rooms" "bathroom" "exterior" "amenities" "pool" "other"	Gallery category
uploadedBy	String	Admin user ID
createdAt	Date	Creation timestamp



API Routes

Public Routes

Method	Endpoint	Description	Auth
GET	/api/rooms	List rooms with filters (check-in/out, featured, pagination)	No
GET	/api/gallery	List gallery images by category	No
POST	/api/contact	Submit contact form	No
POST	/api/bookings	Create new booking	Yes
GET	/api/bookings	List bookings (admin: all, customer: own)	Yes
POST	/api/reviews	Submit room review	Yes

Auth Routes

Method	Endpoint	Description
POST	/api/auth/register	Register new user (with email verification)
POST	/api/auth/forgot-password	Send password reset email
POST	/api/auth/reset-password	Reset password with token
GET	/api/auth/verify	Verify email with token
GET/POST	/api/auth/[...nextauth]	NextAuth.js authentication

Admin Routes

Method	Endpoint	Description
GET	/api/admin/dashboard	Dashboard analytics (revenue, occupancy, bookings)
GET	/api/admin/stats	Basic statistics
GET	/api/admin/rooms	List all rooms (paginated)
POST	/api/admin/rooms	Create room

Method	Endpoint	Description
GET/PUT/PATCH/DELETE	/api/admin/rooms/[id]	Room CRUD operations
GET	/api/admin/rooms/stats	Room statistics
GET	/api/admin/gallery	List gallery images
POST	/api/admin/gallery	Upload gallery image
PATCH/DELETE	/api/admin/gallery/[id]	Update/delete gallery image
POST	/api/admin/upload	Upload image to Cloudinary
GET	/api/admin/users	List all users
PATCH/DELETE	/api/admin/users/[id]	Update/delete user
POST	/api/admin/verify-user	Manually verify user
PATCH	/api/admin/bookings/[id]	Update booking status (confirm/cancel)

Design & User Experience

Visual Design

- **Luxury Aesthetic** — Teal/emerald gradient palette with Playfair Display and Inter fonts
- **Glassmorphism** — Frosted glass effects throughout the interface
- **Smooth Animations** — Framer Motion powered transitions, hover effects, and loading states
- **Responsive** — Mobile-first design with adaptive layouts for all screen sizes
- **Accessibility** — ARIA labels, keyboard navigation, semantic HTML

Performance

- **Turbopack** — Next.js 16's build system for faster compilation
- **Image Optimization** — Next.js Image component with lazy loading
- **Dynamic Imports** — Code splitting for better initial load
- **Edge Caching** — Vercel edge network for global distribution
- **Optimized Fonts** — next/font for zero-layout-shift font loading

SEO & Metadata

- **Open Graph** — Social media sharing cards
- **Twitter Cards** — Twitter sharing support
- **Canonical URLs** — Proper URL canonicalization
- **Robots.txt** — Search engine crawling configuration
- **Manifest** — PWA-ready manifest file

Email System

The project includes a sophisticated HTML email system with:

- **Booking Confirmations** — Professional emails with room details, amenities, and booking summary
- **Booking Cancellations** — Cancellation notifications with rebooking links
- **Welcome Emails** — New account creation emails
- **Welcome Back Emails** — Returning user login emails
- **Password Reset** — Secure reset links with expiry
- **Email Verification** — Account activation emails
- **Contact Form** — Admin notification for inquiries

All emails use a custom-designed template with luxury branding, responsive design, and consistent styling.

Google Analytics 4 Integration

The project implements comprehensive GA4 tracking:

Event	Trigger
view_item_list	Room listing page
view_item	Room detail page
select_item	Room card click
begin_checkout	Booking page load
search_availability	Room search

Event	Trigger
<code>select_date</code>	Date selection
<code>add_guest</code>	Guest count change
<code>view_gallery</code>	Gallery page view
<code>view_reviews</code>	Reviews section view
<code>generate_lead</code>	Contact form submission
<code>login</code>	User login (with method)
<code>user_identified</code>	User session start
<code>video_engagement</code>	Hero video interactions
<code>click_to_call</code>	Phone number click
<code>click_email</code>	Email link click
<code>click_map</code>	Map interaction



Security Features

- **Rate Limiting** — Auth endpoints protected against brute force
- **Bcrypt Hashing** — Secure password storage (12 rounds)
- **JWT Sessions** — Stateless authentication with NextAuth.js
- **Input Validation** — Server-side validation for all forms
- **XSS Protection** — React's built-in escaping
- **CSRF Protection** — NextAuth.js CSRF tokens
- **Secure Headers** — Vercel security headers
- **Environment Variables** — Secrets stored securely



Deployment

Vercel Deployment

The project is deployed on Vercel at zainserenity.vercel.app.

Steps to deploy:

1. Push code to GitHub repository
2. Import project in Vercel
3. Configure environment variables in Vercel dashboard
4. Deploy

Environment Variables on Vercel

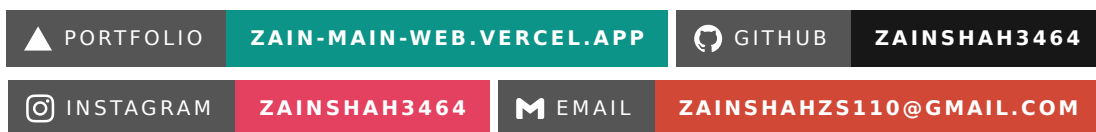
All environment variables are configured in the Vercel dashboard under **Project Settings** → **Environment Variables**. Production and Preview environments share the same variables.



Developer

Zain Shah

Full-Stack Developer & Founder



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- **Framer Motion** — For animation library
- **Tailwind CSS** — For utility-first CSS framework

Built with ❤️ by Zain Shah

"This place is not just my life's work; it's a love letter to the sea, the mountains, and every traveller who dares to dream."